finally told me that Wheeler's guess was wrong. No matter how hard one might squeeze it, Melvin's cylindrical bundle of magnetic field lines will always spring back. Gravity can never overcome the field's repulsive pressure. There is no implosion.

This was the best possible result, Wheeler explained to me enthusiastically: When a calculation confirms one's expectations, one merely firms up a bit one's intuitive understanding of the laws of physics. But when a calculation contradicts expectations, one is on the way toward new insight.

The contrast between a spherical star and Melvin's cylindrical bundle of magnetic field lines was extreme, Wheeler and I realized: When a spherical star is very compact, gravity inside it overwhelms any and all internal pressure that the star can muster. *The implosion of massive, spherical stars is compulsory* (Chapter 5). By contrast, regardless of how hard one squeezes a cylindrical bundle of magnetic field lines, regardless of how compact one makes the bundle's circular cross section (Figure 7.1d), the bundle's pressure will always overcome gravity and push the field lines back outward. *The implosion of cylindrical magnetic field lines is forbidden;* it can never occur.

Why do spherical stars and a cylindrical magnetic field behave so differently? Wheeler encouraged me to probe this question from every possible direction; the answer might bring deep insight into the laws of physics. But he did not tell me how to probe. I was becoming an independent researcher; it would be best, he believed, for me to develop my own research strategy without further guidance from him. Independence breeds strength.

From 1963 to 1972, through most of the golden age, I struggled to understand the contrast between spherical stars and cylindrical magnetic fields, but only in fits and starts. The question was deep and difficult, and there were other, easier issues to study with most of my effort: the pulsations of stars, the gravitational waves that stars should emit when they pulsate, the effects of spacetime curvature on huge clusters of stars and on their implosion. Amidst those studies, once or twice a year I would pull from my desk drawer the stacks of manila folders containing my magnetic field calculations. Gradually I augmented those calculations with computations of other idealized infinitely long, cylindrical objects: cylindrical "stars" made of hot gas, cylindrical clouds of dust that implode,